A CONVERSE JENSEN-MEASURE CRITERION

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Problem 1 (Cole–Gamelin [1, §5, after Lemma 5.1, p. 119])**.** Let A be a uniform algebra on a compact Hausdorff space X, let $\phi \in M_A$, and let μ be a regular Borel probability measure on X. Write $J_\phi(X)$ for the Jensen probability measures for ϕ supported on X, and τ_a for the component of τ absolutely continuous with respect to μ. Characterize

$$\lambda = \sup_{\tau \in J_\phi(X)} \tau_a(X)$$

in terms of A-subharmonic functions. Cole–Gamelin [1, Lemma 5.1] proves $\lambda \le c$ if $u_n \le 0$ on M_A, $\limsup_n u_n \le -1$ μ-almost everywhere, and $u_n(\phi) \to -c$, where $c > 0$.

Theorem 1. *There are continuous A-subharmonic functions u_n on M_A such that*

$$u_n \le 0 \quad \text{on } M_A, \qquad u_n(\phi) \longrightarrow -\lambda,$$
$$\limsup_n u_n(x) \le -1 \quad \text{for } \mu\text{-almost every } x \in X.$$

Thus λ is the minimum of the admissible limits $c \ge 0$ in the criterion above, even when one restricts to continuous functions. It is also the infimum when one requires $c > 0$.

Identify X with its evaluation characters in M_A and write $\widehat{f}$ for the Gelfand transform of $f \in A$. Let $\mathcal{S}$ be the cone of finite positive linear combinations of

$$\ell_{f,N} = \max\{\log|\widehat{f}|, -N\}, \qquad f \in A, \quad N = 1, 2, \dots,$$

together with arbitrary real constants. These are continuous on M_A. The definition in Gamelin [2, Chapter 5, Quasi-subharmonic Functions] requires upper semicontinuity and the Jensen inequality for Jensen measures supported on compact subsets of the domain. Every $\ell_{f,N}$ satisfies that inequality: when $\log|\widehat{f}(\chi)| \ge -N$, use the log Jensen inequality; otherwise its integral is at least $-N$. Positive sums and constants preserve the inequality, so every element of $\mathcal{S}$ is A-subharmonic in precisely that definition.

Moreover, a probability measure τ on X is in $J_\chi(X)$ if and only if

$$u(\chi) \le \int_X u \, \mathrm{d}\tau \quad \text{for every } u \in \mathcal{S}. \tag{1}$$

One direction was just proved. For the other, let $N \to \infty$ in the tests $\ell_{f,N}$, using monotone convergence after subtracting a common upper bound, to recover the log Jensen inequality. Consequently, $J_\chi(X)$ is weak-* closed in the compact space of probability measures, and is convex. It is nonempty: apply Alexander [3, Theorem 1.1] to the multiplicative semigroup $\widehat{A}$ on M_A, boundary X, and probability measure δ_χ. Indeed, $\sup_{M_A}|\widehat{f}| = \sup_X |f|$, since characters are contractive and $X \subset M_A$. This checks the boundary hypothesis for every χ.

Lemma 1. *For every $h \in C(X, \mathbb{R})$,*

$$P(h) := \inf_{v \in -\mathcal{S},\ v|_X \ge h} v(\phi) = \sup_{\tau \in J_\phi(X)} \int_X h \, \mathrm{d}\tau. \tag{2}$$

If $h \ge 0$, every majorant v in this infimum is nonnegative on all of M_A.

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Proof. Constants show that the infimum is finite above. A Jensen measure at ϕ gives $v(\phi) \geq \int v \,\mathrm{d}\tau \geq \min_X h$, so it is finite below. The functional P is sublinear, monotone, and satisfies $P(h+c) = P(h)+c$ for real constants c. Fix h_0. The real Hahn–Banach theorem gives a linear functional $L \leq P$ with $L(h_0) = P(h_0)$. For $g \geq 0$, $L(-g) \leq P(-g) \leq 0$; hence L is positive. Since $P(c) = c$, $L(1) = 1$. By the Riesz representation theorem, $L(h) = \int h \,\mathrm{d}\tau$ for a regular Borel probability measure τ. For $v \in -\mathcal{S}$,

$$\int_X v \,\mathrm{d}\tau = L(v|_X) \leq P(v|_X) \leq v(\phi),$$

so Equation (1) implies $\tau \in J_\phi(X)$. This proves the nontrivial inequality in Equation (2); the other follows immediately from Jensen's inequality. Finally, for any $\chi \in M_A$ choose $\sigma \in J_\chi(X)$. If $v|_X \geq h \geq 0$, then $v(\chi) \geq \int_X v \,\mathrm{d}\sigma \geq 0$. □

Lemma 2. *For $M > 0$, put*

$$\lambda_M = \sup_{\tau \in J_\phi(X)} (\tau \wedge M\mu)(X),$$

where the meet of two positive measures has density equal to the minimum of their densities relative to their sum. Then

$$\lambda_M = \inf_{\substack{h \in C(X,\mathbb{R}) \\ 0 \leq h \leq 1}} \left(P(h) + M \int_X (1-h) \,\mathrm{d}\mu \right), \qquad \lambda_M \uparrow \lambda \quad (M \to \infty). \tag{3}$$

Proof. For fixed τ, with $\nu = \tau + M\mu$, write $a = \mathrm{d}\tau/\,\mathrm{d}\nu$ and $b = M\,\mathrm{d}\mu/\,\mathrm{d}\nu$. Pointwise minimization gives

$$(\tau \wedge M\mu)(X) = \inf_{\substack{h \in C(X,\mathbb{R}) \\ 0 \leq h \leq 1}} \left(\int_X h \,\mathrm{d}\tau + M \int_X (1-h) \,\mathrm{d}\mu \right).$$

To justify continuous rather than measurable tests, approximate the minimizing indicator $1_{\{a<b\}}$ in $L^1(\nu)$ by continuous functions between zero and one. This follows from inner and outer regularity and Urysohn's lemma on compact Hausdorff spaces. The displayed objective changes by at most the $L^1(\nu)$ approximation error. Apply Sion [4, Theorem 4.2] to $J_\phi(X)$ and the convex set $\{h \in C(X,\mathbb{R}) : 0 \leq h \leq 1\}$. The first set is compact; the objective is affine in both variables and weak-* continuous in τ for fixed h. The theorem interchanges the supremum and infimum. Then Equation (2) gives Equation (3). If $\tau = f_\tau \mu + \tau_s$, its overlap mass equals $\int_X \min(f_\tau, M) \,\mathrm{d}\mu$, and increases to $\tau_a(X)$. Taking suprema in either order over M and τ proves the last assertion. □

Proof of Theorem 1. Let $M_n = 2^n$ and $\varepsilon_n = 2^{-n}$. Choose $0 \leq h_n \leq 1$ continuous with

$$P(h_n) + M_n \int_X (1-h_n) \,\mathrm{d}\mu \leq \lambda_{M_n} + \varepsilon_n,$$

and choose $v_n \in -\mathcal{S}$, $v_n|_X \geq h_n$, such that $P(h_n) \leq v_n(\phi) \leq P(h_n) + \varepsilon_n$. By Lemma 1, $v_n \geq 0$ on M_A and $P(h_n) \geq 0$. Therefore

$$0 \leq v_n(\phi) \leq \lambda + 2\varepsilon_n,$$

$$\int_X (1-v_n)_+ \,\mathrm{d}\mu \leq \int_X (1-h_n) \,\mathrm{d}\mu \leq \frac{\lambda + \varepsilon_n}{M_n}.$$

The last quantities are summable. Tonelli's theorem implies $\sum_n (1-v_n(x))_+ < \infty$ for μ-almost every x; in particular, $\liminf_n v_n(x) \geq 1$ there. For any $\tau \in J_\phi(X)$, nonnegativity and Jensen's inequality give

$$v_n(\phi) \geq \int_X v_n \,\mathrm{d}\tau \geq \int_X v_n \,\mathrm{d}\tau_a.$$

Fatou's lemma, with $\tau_a \ll \mu$, yields $\liminf_n v_n(\phi) \geq \tau_a(X)$. Taking the supremum over τ and using the upper bound proves $v_n(\phi) \to \lambda$ along the entire sequence, including when $\lambda = 0$ or 1. Set $u_n = -v_n$. These belong to $\mathcal{S}$ and have all the required properties.

Conversely, for any admissible A-subharmonic sequence and $c \geq 0$, put $w_n = -u_n \geq 0$. The same Jensen–Fatou argument gives $c \geq \tau_a(X)$ for every τ, hence $c \geq \lambda$. The Jensen inequality for the source's full subharmonic class is stated at the start of Cole–Gamelin [1, §5]; the argument also proves its Lemma 5.1 at the omitted degenerate value $c = 0$. Finally, replacing u_n by $u_n - \delta$ for any $\delta > 0$ supplies an admissible positive limit $c = \lambda + \delta$, proving the infimum claim. □

Remark 1. The source does not require Jensen measures to be supported on X. Nothing changes if λ is defined over all Jensen measures τ for ϕ on M_A, with τ_a still taken relative to μ. Indeed, the functions u_n of Theorem 1 satisfy the Jensen inequality for such τ, so the Fatou argument above bounds $\tau_a(X)$ by the value λ defined over $J_\phi(X)$. The reverse inequality is trivial.

Remark 2. This is a full converse for arbitrary compact Hausdorff X; it assumes neither metrizability nor the infinite-polydisk setting. The argument proves the cone duality, global nonnegativity, and the almost-everywhere sequence construction. The duality mechanism is classical; no novelty claim is made for it.

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